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*3. The average kinetic energy of one monatomic gas molecule at temperature T is given by

$$E_K = \frac{3}{2} \left(\frac{R}{N_A} \right) T,$$

where R is the universal gas constant and N_A is the Avogadro constant. A monatomic gas is heated from 300 K to 350 K under fixed volume.

(a) Estimate the ratio of the root-mean-square speed ($c_{r.m.s.}$) of the gas molecules at the two temperatures

$$\left(\frac{c_{r.m.s.} \text{ at } 350 \text{ K}}{c_{r.m.s.} \text{ at } 300 \text{ K}} \right).$$

(2 marks)

(b) Thus, using kinetic theory, explain why the gas pressure would increase.

(2 marks)

Answers written in the margins will not be marked.

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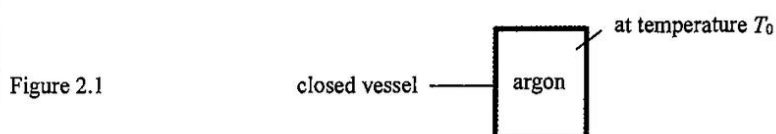
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Book 1 section 4.3: DSE long questions

2017 Q3 (page 2)

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- *2. Figure 2.1 shows a closed vessel of a fixed volume containing argon, a monatomic gas, maintained at temperature T_0 . The root-mean-square speed ($c_{\text{r.m.s.}}$) of the argon atoms inside is 500 m s^{-1} .



- (a) (i) Calculate the average kinetic energy of the argon atoms, E_K . Given that the mass of an argon atom is $6.63 \times 10^{-26} \text{ kg}$. (2 marks)

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- (ii) Hence, or otherwise, find T_0 (in K). (2 marks)

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- (b) If some of the argon has been leaked from the vessel while the temperature remains at T_0 , state and explain the change, if any, in the root-mean-square speed of the argon atoms inside the vessel. (2 marks)

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